

NAT AND PAT CONCEPTS

Question 1

In Cisco Packet Tracer, set up a simple network with a router, one internal PC, and one external PC. Assign private IP addresses to the internal PC and public IP addresses to the external PC, then configure static NAT on the router to allow the internal PC to communicate with the external PC.

Answer: NAT (Network Address Translation) converts a private IP address into a public IP address so an internal PC can communicate with an external network.

Device	IP Address
Internal PC	192.168.1.10
Router Inside	192.168.1.1
Router Public	203.0.113.1
External PC	203.0.113.10

Static NAT command:

```
ip nat inside source static 192.168.1.10 203.0.113.10
```

NAT / PAT - PACKET TRACER TOPOLOGY

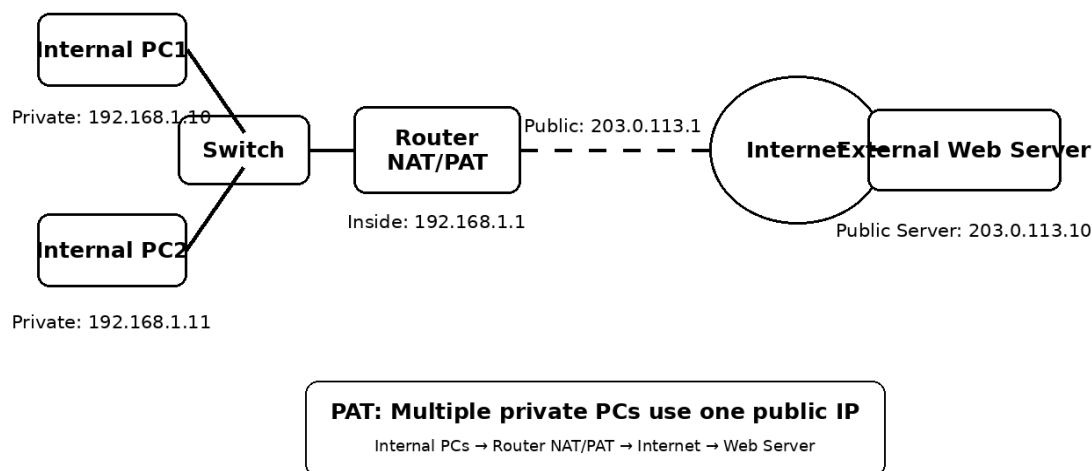


Figure 1: NAT/PAT topology

Question 2

Configure Port Address Translation (PAT) on a router in Cisco Packet Tracer so that two internal PCs with private IPs can access a simulated public web server using a single public IP address.

Answer: PAT allows multiple private PCs to use one public IP address to access the external network.

```
access-list 1 permit 192.168.1.0 0.0.0.255  
ip nat inside source list 1 interface g0/1 overload
```

The word **overload** enables PAT.

Question 3

Test your NAT configuration by sending a ping from the internal PC to the external PC and capture the NAT translation table on the router to verify the address mapping.

Answer: Test connectivity from the internal PC:

```
ping 203.0.113.10
```

Check the NAT translation table:

```
show ip nat translations
```

The table shows the private-to-public IP translation.

NAT Translation Table - Example

Type	Inside Local	Inside Global	Destination
PAT	192.168.1.10:1025	203.0.113.1:30001	203.0.113.10
PAT	192.168.1.11:1026	203.0.113.1:30002	203.0.113.10
PAT	192.168.1.12:1027	203.0.113.1:30003	203.0.113.10

Multiple private devices share one public IP using different port numbers.

Figure 2: Example NAT translation table

Question 4

Modify your Packet Tracer topology to add a third internal PC. Update your NAT or PAT configuration so all three internal PCs can access the external network, and verify that each translation appears in the router's NAT table.

Answer: Add a third internal PC with:

```
PC1 → 192.168.1.10
PC2 → 192.168.1.11
PC3 → 192.168.1.12
```

All three PCs can use the same public IP through PAT. Check the translations with:

```
show ip nat translations
```

NAT Translation Table - Example

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PAT	192.168.1.10:1025	203.0.113.1:30001	203.0.113.10
PAT	192.168.1.11:1026	203.0.113.1:30002	203.0.113.10
PAT	192.168.1.12:1027	203.0.113.1:30003	203.0.113.10

Multiple private devices share one public IP using different port numbers.

Figure 3: Example showing multiple PAT translations

Question 5

Explain in your own words how NAT and PAT help apps like Zomato or Flipkart serve millions of users with limited public IP addresses. Give two real-world scenarios where incorrect NAT/PAT setup could break connectivity for users.

Answer: NAT converts private IP addresses into public IP addresses, while PAT allows many devices to share one public IP address. This helps services like Zomato and Flipkart support many users while saving public IPv4 addresses.

Two real-world problems:

1. Wrong NAT configuration can stop users from accessing external servers.
2. Wrong PAT configuration can cause multiple users to lose Internet connectivity.

Conclusion: NAT = Private IP → Public IP. PAT = Multiple Private IPs → One Public IP.